

Supplemental material for XYZ

1 SUMMA details

At CSSL, the peak water equivalent was XYZ higher in the open site relative to the forest site leading up to the heatwave event which may have resulted from a combination of snow interception or antecedent melt. We found good correspondence between SUMMA and observations in the open and forest sites, respectively. This was found after adjusting the rain/snow threshold temperature to XYZ, the canopy wind reduction coefficient to and a snow surface roughness of 0.002 m for both sites. These values are well within established bounds for those parameters (CITE). Comparisons against the forest site CNR4 radiometer showed good correspondence between the SUMMA modelled below-canopy shortwave radiation, lending confidence to the prescribed LAI value and UEB 2-stream shortwave parameterization implemented in SUMMA.

2 Cirrus and RRTMG

Then, we insert a hypothetical cirrus cloud at 200 hPa in the model with a cloud water path of 6 gmtwo and an effective cloud radii of todo typical, for mid-latitude cirrus cloud todo $cCRF_x$ is computed as the difference between the four component net radiative difference between the cloudy atmosphere simulation and the clear-sky simulation.

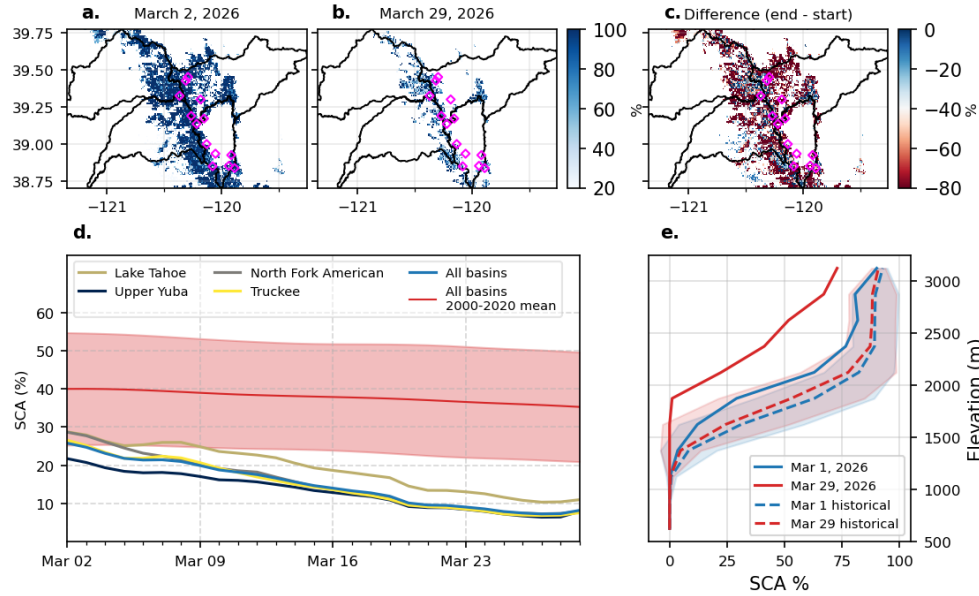


Figure S1: SPIRES SCA across the study region during the March heatwave. Watershed boundaries shown in black and SNOTEL monitoring sites indicated by magenta diamonds. **(a, b, c)** SCA (%) on March 2 and March 29, 2026, respectively, and the difference between the two dates. **(d)** Daily mean SCA (%) for each individual watershed and aggregated across all four basins. The 2000–2020 historical mean and ± 1 standard deviation are depicted. **(e)** Mean SCA as a function of elevation band (250 m bins) for March 2 (blue) and March 29 (red), shown alongside historical values for the same dates (dashed lines) with ± 1 standard deviation shading.